

Rebound nystagmus, a window into the oculomotor integrator

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Abstract

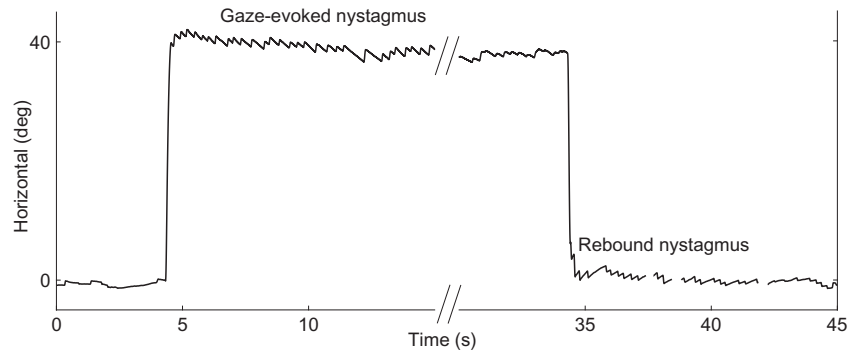
Rebound nystagmus, a common cerebellar sign, is a transient nystagmus that appears on returning to straight-ahead gaze after prolonged eccentric gaze. The slow phases of rebound nystagmus are in the direction of prior eccentric gaze. After eccentric gaze, healthy subjects also show rebound nystagmus when fixation is removed. Rebound nystagmus is thought to be related to the function of the oculomotor neural integrator—the circuit that ensures accurate gaze holding after any eye movement—but the exact mechanism of rebound nystagmus is unknown. Here, we combine experimental data with mathematical modeling to test several hypotheses for the generation of rebound nystagmus. We show that two mechanisms contribute, one relies on vision and the other does not. Future experiments must determine if (1) the non-visual mechanism is related to eye position or to eye velocity signals and (2) whether these signals are based on afferent (proprioception) or efferent (corollary) information.

Keywords

Cerebellum, Drift, Gaze-evoked nystagmus

1 Introduction

Rebound nystagmus (RN) is a transient nystagmus that occurs when the eyes return to straight ahead position following a period of prolonged eccentric gaze holding (Fig. 1). In RN the slow phases are directed toward the former eccentric gaze position and the quick-phases away from it (Leigh and Zee, 2015). Hood first described RN in

**FIG. 1**

Example of rebound nystagmus. A healthy subject looks at a flashing target, first at center (i.e., straight ahead), then eccentrically at 40 degrees to the right for 30s, and then back to the center. During eccentric fixation at 40 degrees there is a gaze-evoked nystagmus with the slow phase drifting toward the center, which gradually decreases in intensity. During the rebound period, the slow phase drifts toward the previously-held, eccentric position.

a group of patients with cerebellar disorders (Hood et al., 1973) and it is typically considered to be a sign of abnormal cerebellar function (Lin and Young, 1999; Sharpe, 1974; Zee et al., 1976). Less intense RN can also be found in healthy subjects (Gordon et al., 1986; Shallo-Hoffmann et al., 1990; Zee et al., 1976), especially in the absence of a fixation target upon return of gaze to center.

Stable gaze holding is achieved by a neural network within the brainstem and cerebellum. This neural network must integrate (in the mathematical sense) signals that encode the velocity of eye movement commands such as saccades, smooth pursuit, or the vestibular ocular reflex, into position commands for holding the eyes still. The motor neurons receive the direct command encoding the velocity of the desired movement and the position command from the output of the neural integrator. These two signals combine in what is known, for saccades, as the pulse-step signal. The pulse helps overcome the viscous drag of the tissues surrounding the eye and the step provides the sustained drive to hold the eye at its new position (Robinson, 1981).

The neural integrator for gaze holding is not perfect, and even in some normal subjects, the eyes drift centripetally after an eccentric saccade, i.e., toward the straight-ahead position of gaze. This centripetal drift causes a nystagmus that is commonly referred to as gaze-evoked nystagmus. Gaze-evoked nystagmus tends to be larger at more eccentric eye positions and there is a null position where there is no drift and where the eye drifts toward from other positions. If we assume the drift can be approximated with an exponential curve, it can be parameterized by its time constant. In this case, the combination of time constant and null position determines the velocity of the drift from a given eye position.

The existence of rebound nystagmus indicates that the properties of the integrator may not be constant over time and instead depend on the prior eye movements and eye positions. If patients with gaze-evoked nystagmus are encouraged to sustain their attempt to look eccentrically, their nystagmus may quiet down and, on some occasions, even reverse direction, so that the eyes begin to drift centrifugally (Hood et al., 1973; Leech et al., 1977; Zee et al., 1976). This adaptability to past ocular motor behavior may represent mechanisms to optimize the behavior of the integrator. Understanding these mechanisms may help interpret the rebound nystagmus present in patients and improved their diagnosis and treatment.

In this study, we examined rebound nystagmus in healthy subjects with a briefly flashing target (to eliminate image motion on the retina due to the nystagmus itself) vs. continuously illuminated targets and tested whether bringing the eye to eccentric positions with different types of eye movements (smooth pursuit or saccades) affects either the intensity or duration of the rebound nystagmus. We also implemented a computational model to simulate the current results and make predictions of the results of future experiments to test the possible mechanisms of rebound nystagmus.

2 Methods

Six healthy human volunteers (two women) participated in the study. The experimental procedures were approved by Johns Hopkins Institutional Review Board and written informed consent was obtained from each subject. Subjects had no history of neurological nor vestibular disorders and were not taking sedative or psychiatric medications.

All experiments were performed in a dark room. Subjects sat upright with the head immobilized by a dental bite bar 135 cm away from a light-transmissive horizontal screen. Red laser targets were rear-projected on the screen and moved by a mirror galvanometer. Horizontal eye movements were recorded with an infrared video goggle system (RealEyes xDVR system, Micromedical Technologies Inc., Chatham, IL) controlled by custom software (Otero-Millan et al., 2015) at a frame rate of 100 Hz. Each subject participated in two experiments.

Experiment I measured the difference on rebound nystagmus between holding eccentric gaze with a flashing target and with a continuous target. Each subject was asked to fix on a flashing target (illuminated for 10 ms every second) in the straight-ahead position. After 5 s, the target jumped eccentrically ($\pm 40^\circ$) and either flashed or remained on continuously. After 30 s holding the eccentric gaze position, the target jumped back to straight-ahead position and flashed (10 ms on per second) for 15 s. After the trial finished subjects had a 10-s break before the next trial began. For each condition (flashing or continuous target), two blocks of five trials were performed, one for each direction (left or right).

Experiment II measured the difference in rebound nystagmus after bringing the eye to an eccentric position by different types of eye movements. Three experimental conditions were used: single saccade, smooth pursuit, and a stair-case of saccades. For all conditions, the target was flashing at the straight-ahead position and at the 40° position but continuously on during the movements toward the eccentric target. For single saccade trials, we used the data from Experiment I where after 5 s recordings in the straight-ahead position, the subjects were asked to make a saccade to an eccentricity of 40° . After 30 s the target jumped back to the straight-ahead position where it remained for another 15 s. For smooth pursuit movements, after 5 s recordings in the straight-ahead position, the subjects were asked to track a small red laser dot, moving at $4^\circ/\text{s}$ to reach an eccentricity of 40° which took 10 s. After a further 20 s of eccentric fixation, the target jumped back to the straight-ahead position where it remained for another 15 s. For staircase saccades, after 5 s with flashing light in the straight-ahead position, the target jumped four degrees every second until reaching an eccentricity of 40° (a total of 10 saccades which took 10 s.). After a further 20 s of eccentric fixation the target jumped back to the straight-ahead position where it remained for another 15 s. Note that in all cases the time spent away from the straight-ahead position was 30 s. For each condition, two blocks of five trials were performed, one for each direction (left or right).

To quantify rebound nystagmus, we measured the velocity of the slow phase of the nystagmus over time as the median velocity within windows of 500 ms (excluding samples belonging to quick phases, identified with a velocity threshold of $10^\circ/\text{s}$ in either direction). Then we fit an exponential curve to the data corresponding to the first 15 s after returning to the straight-ahead position. From the exponential fit we obtained the maximum slow-phase velocity of the nystagmus and the time constant of the decay of the slow-phase velocity over time.

3 Results

In Experiment I we found that rebound nystagmus was more pronounced after subjects fixated eccentrically at the continuously illuminated target than at the flashing target. The average maximum slow-phase velocity, as measured with an exponential fit, was $3.5 \pm 0.7^\circ/\text{s}$ for the continuous target and $2.3 \pm 0.5^\circ/\text{s}$ for the flashing target ($P=0.036$ paired t -test). The average time constant was 9 ± 3 s for the continuous target and 6 ± 1 s for the flashing target ($P=0.1$ paired t -test). In Experiment II we found that the velocity of the rebound nystagmus did not depend on how the eyes reached the peripheral target: with a single saccade, 10 small saccades, or smooth pursuit. The average maximum slow-phase velocity was $2.3 \pm 0.5^\circ/\text{s}$, $1.9 \pm 0.4^\circ/\text{s}$, and $1.6 \pm 0.2^\circ/\text{s}$ for each of the conditions, respectively ($P=0.2$ repeated measures anova). The time constant of the rebound nystagmus was 6 ± 1 s, 6 ± 2 s, and 6 ± 2 s, respectively ($P=0.7$ repeated measures anova). Fig. 2 shows the average slow-phase velocity across subjects as well as the individual subjects for each of the experiments for the first 15 s after the subjects returned to straight ahead position.

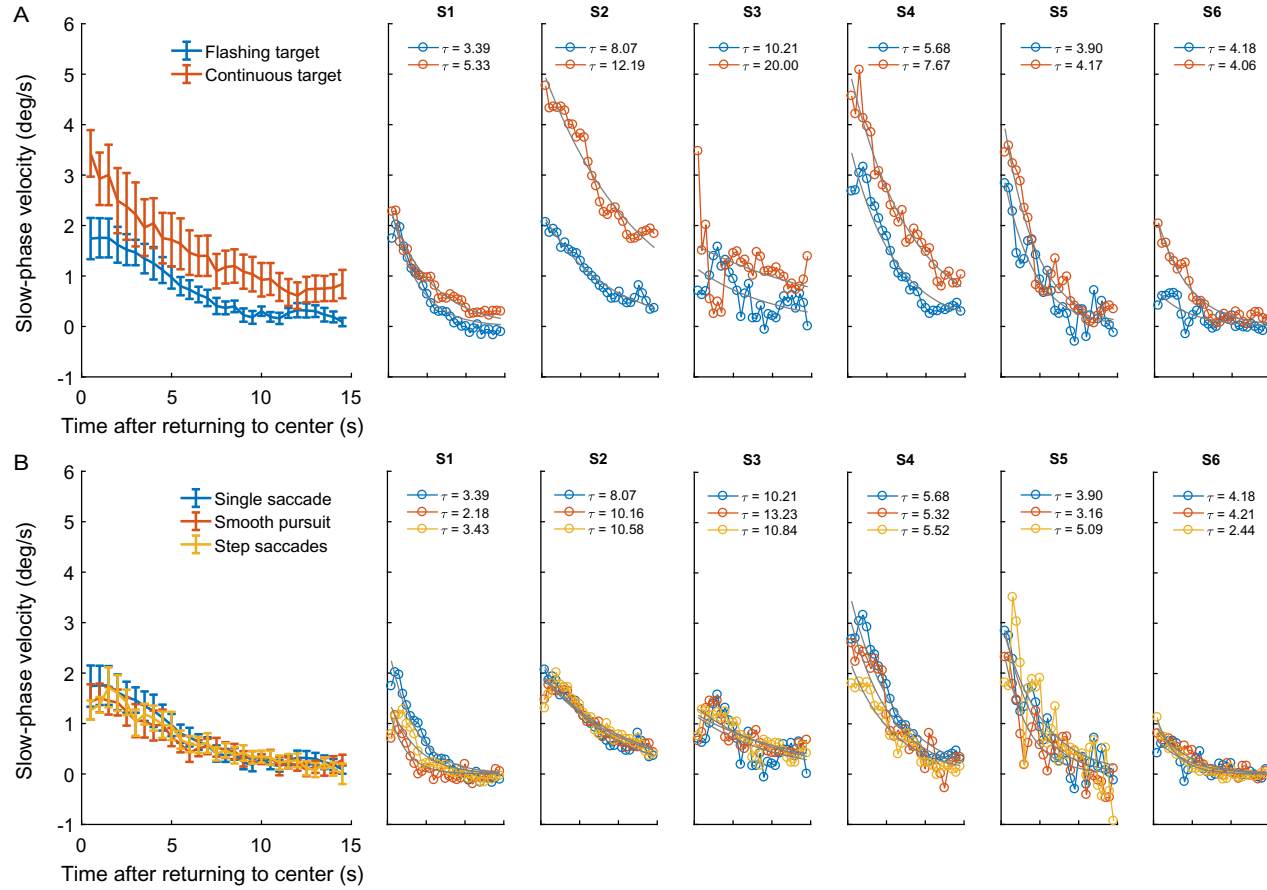


FIG. 2

Results of Experiments I (A) and Experiment II (B). The first columns show the average across subjects with error bars displaying the standard error of the mean. Subsequent columns show the results for each individual subject.

4 Discussion

4.1 Characterization of gaze holding

Gaze-holding behavior can be characterized by measuring the velocity of the gaze-evoked nystagmus as a function of the eye position eccentricity. This relationship is usually not linear (Abel et al., 1978; Bertolini et al., 2013) and can be characterized with a tangent function (Bertolini et al., 2013). Specifically, the relationship between eccentricity and drift (slow phase) velocity can be described with the simple equation (Romano et al., 2017; Tarnutzer et al., 2015):

$$V_E = \frac{k_2}{k_1} \tan(k_1 * E) + c_2$$

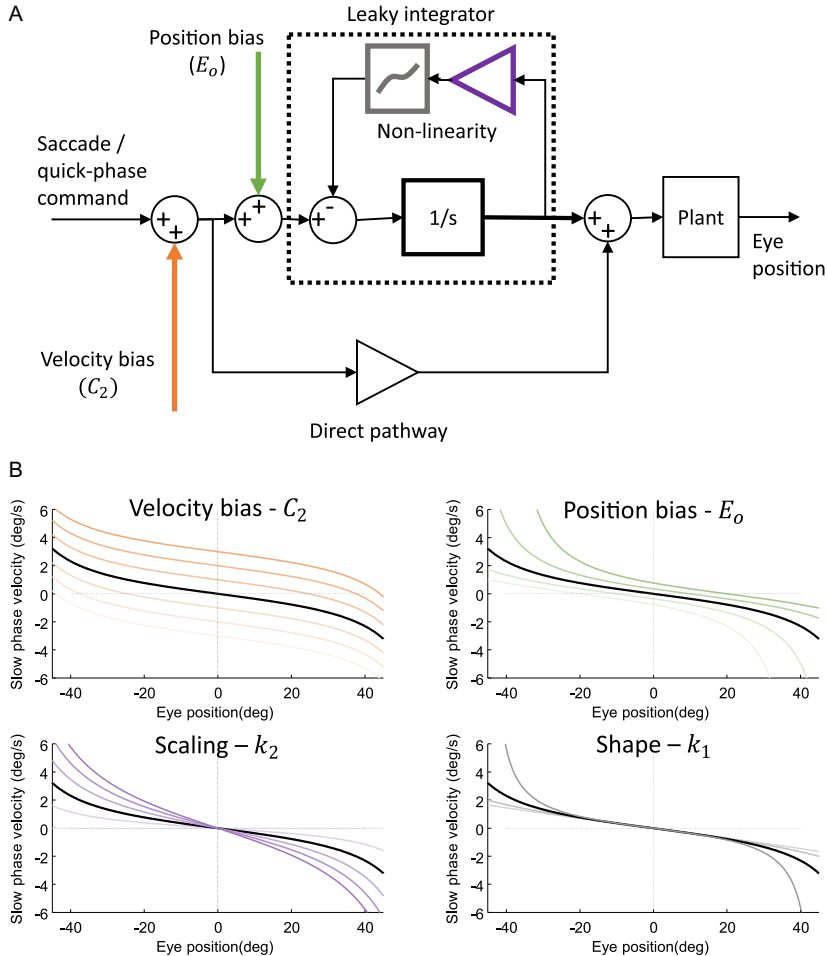
where V_E is the slow-phase velocity of the nystagmus at eccentricity E , k_1 is a parameter that defines the shape of the curve, that is, how non-linear it is, k_2 is a scaling factor that changes the overall slope of the curve, and c_2 is a velocity bias that corresponds with a possible eye drift present at straight ahead position. They used this approach to characterize the behavior of the neural integrator for gaze holding in healthy individuals (Bertolini et al., 2013), patients with different neurological disorders (Tarnutzer et al., 2015), and individuals with alcohol intoxication (Romano et al., 2017).

The parameters that characterize the tangent function need not be stable over time and may be affected by the recent history of eye movements, as is the case in rebound nystagmus. In this context the parameters of the tangent function become functions of time. Additionally, we added another parameter E_o that represents a possible shift in the null position of the integrator over time and moves the curve horizontally.

$$V_E(t) = \frac{k_2(t)}{k_1(t)} \tan(k_1(t) * (E - E_o(t))) + c_2(t)$$

Thus, it is possible that initially E_o and c_2 are zero, resulting in no nystagmus at straight-ahead position. However, after holding gaze eccentrically, those parameters may change resulting in a curve that corresponds with rebound nystagmus at straight-ahead position. The other parameters k_1 and k_2 may also change as a result of holding gaze eccentrically. Their changes, however, would not lead to rebound nystagmus by themselves. Instead, they would interact with the potential changes to E_o and c_2 affecting the amount of rebound nystagmus that would be generated.

This analytical characterization of the integrator can be implemented using a control system approach (Fig. 3). A leaky integrator is the most basic form of a gaze-holding control system. It is a system that accumulates the input it receives, in this case eye velocity, creating a memory of eye position. However, the integrator forgets (leaks) at some rate, which results in the memory of the eye position drifting over time. Two main parameters determine the behavior of the integrator: the null position, that is, the position that the eye tends to drift toward in the absence of an input and the time constant, which determines how fast the eye drifts toward the null

**FIG. 3**

Characterization of gaze-holding behavior. (A) Control systems diagram of the main elements of the gaze-holding circuit. (B) Effect of changing each of the parameters of the tangent equation describing the drift velocity of the eye as a function of the eye position. The elements in (A) that correspond to each of the parameters are highlighted with the same color.

position. The slow-phase velocity is determined by both parameters: the lower the time constant the faster the drift; the farther the null position from the current position the faster the drift. If the time constant does not depend on the eye position, the relationship between eye position and drift velocity is linear. To achieve the tangent curve described previously it is necessary for the time constant of the integrator to decrease with eccentricity as determined by k_1 and k_2 . The parameter E_0 corresponds with the null position of the integrator and the parameter c_2 with a constant-velocity bias that is added to the input or output of the integrator.

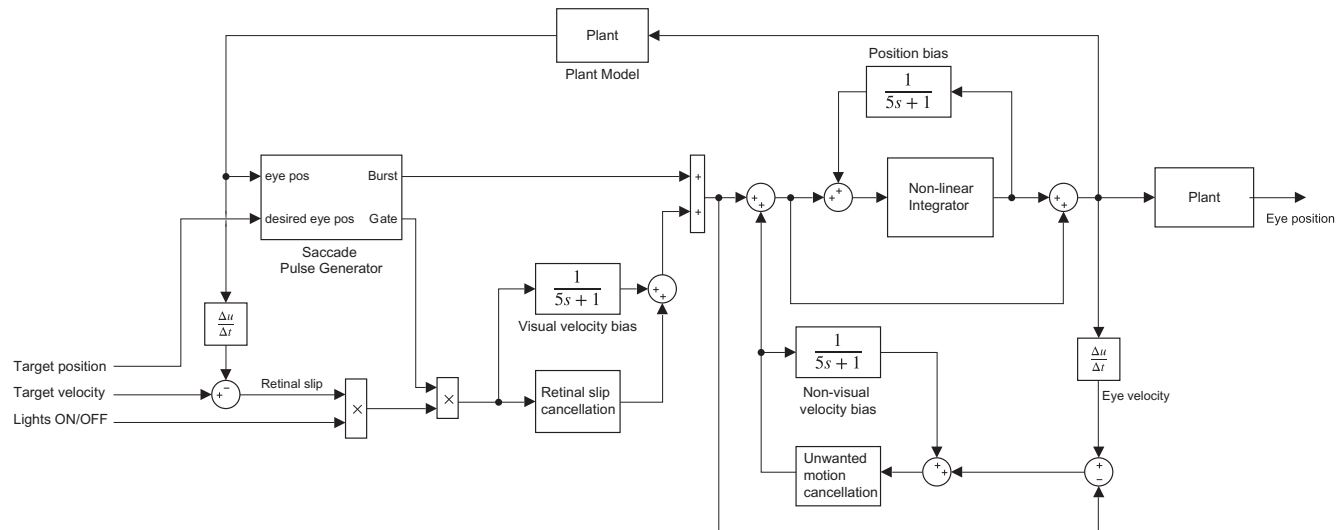


FIG. 4

Diagram of the control system model implemented to simulate rebound nystagmus. Three different elements contribute to rebound nystagmus: position bias, visual velocity bias, and non-visual velocity bias. The non-linear integrator was implemented simply with two different leaks, one for eccentricities within $\pm 25^\circ$ and a larger leak for eccentricities outside of $\pm 25^\circ$. The saccade generator generates a pulse of constant velocity whenever the difference between the target position and the output of the integrator is more than 0.2° until it corrects for that difference. The plant and retinal slip and unwanted motion cancellation systems are modeled as simple first order systems.

4.2 Modeling rebound nystagmus

Here, we set out to implement a control systems model that simulates rebound nystagmus under different conditions, including our results, and can help us design future experiments to test these ideas (Fig. 4). From our results, we know there must be at least two different mechanisms for rebound nystagmus: one independent of vision to account for rebound with the flashing target and one dependent on vision to account for the increased rebound in the continuous target condition. In this model, we implement three mechanisms that may contribute to rebound nystagmus. First, a system like smooth-pursuit, perhaps special for fixation (Luebke and Robinson, 1988), driven by motion of images on the retina (retinal slip) produced by the nystagmus itself, which creates a signal that attempts to cancel the nystagmus. By averaging this signal over time, using a first order system, a velocity bias is introduced. Second, a position bias is generated by averaging the output of the integrator over time and feeding it back to its input. This shifts the null of the integrator toward the current eye position. Finally, a system that may use efference copies of eye movement commands or extraocular proprioception to monitor unwanted drift of the eye also generates a signal that attempts to cancel the nystagmus even in the absence of light. This signal is also averaged over time resulting in another source of velocity bias.

First, we simulated the results of our experiments. Fig. 5A shows the simulation of Experiment I. As in our results, the velocity of the rebound nystagmus is larger in the light condition (equivalent to the continuously on target) than the “dark” condition (equivalent to the flashing target). Fig. 5B shows the simulation of Experiment II. As in the data from our experiments the model produces equivalent velocities of the rebound using smooth pursuit and large or step saccades.

Second, we simulated the results from Chung and Bedell (1995) studying the “dumping” of rebound nystagmus by the presence of a sustained visual target. They found that if a fixation target appeared shortly after returning to the straight-ahead position, the nystagmus would resume after the target disappeared but with a speed lower than would have been without a fixation target. Fig. 5C shows the result of the simulations with a fixation target visible for 4 s. The rebound resumes after the target disappears but the velocity of the nystagmus is lower than for the same timepoint without any visual target having been presented. In this model, the source of this suppression is the velocity bias created by the retinal slip while the fixation target was on.

Finally, we simulated the results of a hypothetical experiment (Fig. 5D and E) where we could measure the contributions of some of the non-visual components. In this experiment, subjects would fix eccentrically for a period of time and then would return not to the straight-ahead position, but to different eye positions covering the entire range of eccentricities. Measuring the velocities of the rebound we could build the curve describing the relationship between eye position and drift velocity to describe how the integrator has changed after holding eccentric gaze. Then, by comparing with the same curve obtained without prior eccentric gaze holding, we could measure the contribution of the different components (position bias and velocity bias) to rebound nystagmus.

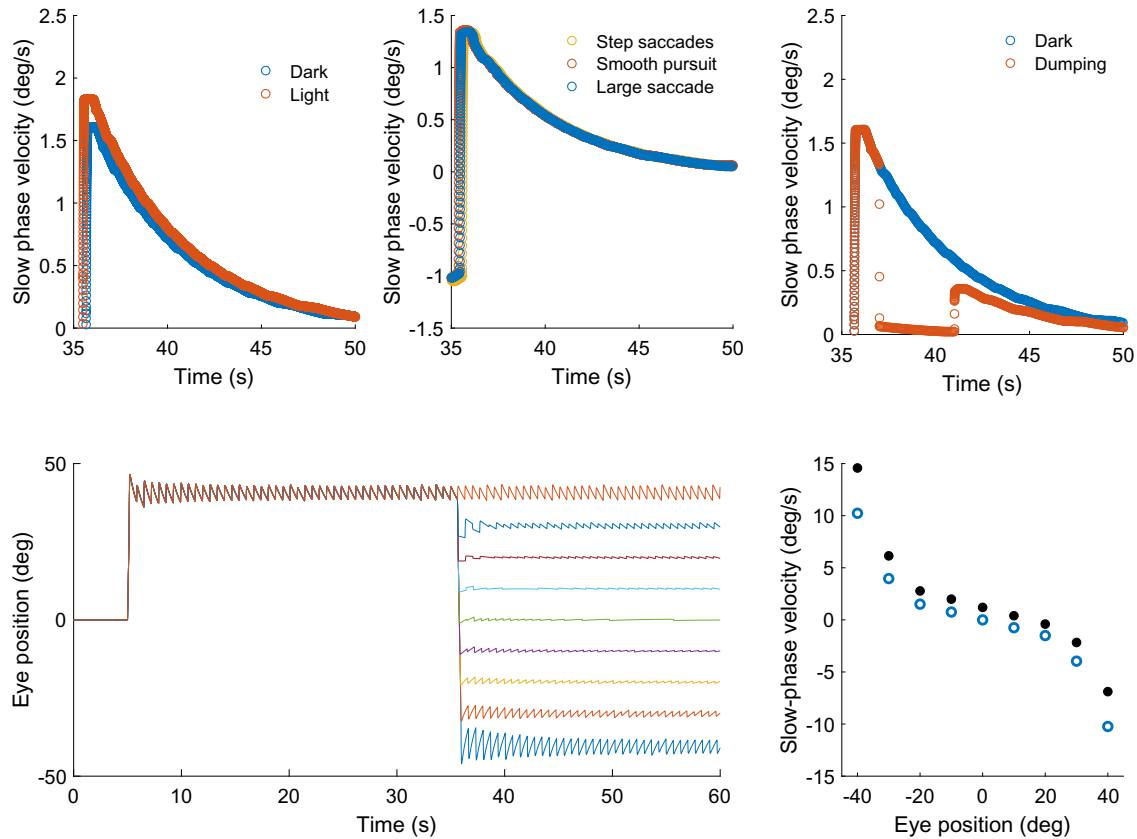


FIG. 5

Results of simulations of the model. (A) Simulations of Experiment I. (B) Simulations of Experiment II. (C) Simulations of [Chung and Bedell \(1995\)](#) experiments. (D and E) Simulations of hypothetical experiment to measure the eye position vs. drift velocity curve during rebound nystagmus. (D) Eye position traces when measuring rebound at each eccentricity after holding gaze always at 40° . (E) Corresponding slow-phase velocities of the rebound nystagmus. Blue dots indicate the velocity of the nystagmus during normal eccentric gaze and black dots represent the velocity of the rebound nystagmus after holding eccentric gaze and then moving to different eye positions.

4.3 Why rebound?

One may ask why the integrator should behave in a way that results in rebound nystagmus. A possible explanation is that there are mechanisms that calibrate the integrator to ensure that most of the time we do not suffer from gaze-evoked nystagmus. This would be part of a general strategy of the brain to eliminate unwanted biases in order to produce “quiet” zones or set-points that allow for optimal sensory and motor behavior (Zee et al., 2017). Based on this idea, the adaptive mechanisms of the brain could keep track of the most common gaze positions, and then change the properties of the integrator to reduce any drift at these positions, but perhaps at the expense of less stability for gaze at uncommon positions. This strategy would correspond with a continuous shift of the null position of the integrator toward the more common eye positions. Indeed, when such a mechanism goes awry, for example, in cerebellar disease, the null position becomes unstable and unwanted drift of the eyes occurs (Leech et al., 1977; Robinson, 1974).

The oculomotor integrator for gaze holding, which also operates upon vestibular signals, would also help compensate for any long-term bias in the vestibular system as another form of set-point adaptation (Khojasteh et al., 2013; Robinson et al., 1984). In this case, there could be a mechanism that continuously monitors the output of the integrator and detects biases in its rate of change (eye velocity). This would correspond with a velocity bias that attempts to compensate and remove unwanted image motion of the retina from instability of gaze. In both cases, the mechanisms must estimate the statistics of signals within the integrator, its inputs, or its outputs. Those estimates must be calculated over a period of time and thus would lag any change of the signals. This lag may not be noticeable under most circumstances. However, if the subject is presented with a sudden and large change, outside of the normal range, the consequences of the lag may be apparent as it is the case in rebound nystagmus.

A slightly different variant on these ideas, which emphasizes the flexibility of these adaptive circuits, is the ability of the brain to change the time constant of the oculomotor integrator to adjust the phase of the vestibulo-ocular reflex to eliminate unwanted retinal image motion during head movements. Normally, such a change in the time constant of the integrator would also lead to unwanted drift of the eyes after every movement ends (Kramer et al., 1995). The adaptive circuits, however, selectively alter integrator function for the vestibular system without causing unwanted drift during fixation after saccades (Kramer et al., 1998).

4.4 Other approaches to model neural integrators

Here, we have focused on a control systems approach to study the gaze-holding integrator and the possible mechanisms responsible for rebound nystagmus. However, there are other models of neural integration (Goldman et al., 2009). For example, Cannon and colleagues (Cannon and Robinson, 1985; Cannon et al., 1983) developed a neural network model that relies on laterally inhibited neurons that

simulate the behavior of neurons identified as part of the neural integrator as well as the expected eye movement behavior. Seung and colleagues developed a model based on a network of neurons with recurrent excitatory connections that behave as an “attractor” (Seung, 1996; Seung et al., 2000). Other models of neural integrators used to account for activity of head direction cells (Song and Wang, 2005) behave as maps where the desired output eye position is encoded by the location of activity in the map. These approaches could also be used to study rebound nystagmus and rebound nystagmus may emerge as an intrinsic property of these implementations. Nonetheless, a simple, control-systems approach as used here accounts for many aspects of rebound nystagmus and suggests future experiments.

Acknowledgments

We thank Dale Roberts for technical assistance, and Jing Tian for her comments. This work was supported by the National Eye Institute (Award K99EY027846 to J.O.M.), the National Institute of Deafness and Other Communication Disorders (NIDCD) (Award 5K23DC013552 to A.K.), and the Leon Levy, Fight for Sight foundations.

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